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Performance enhancement-based active learning sample selection method

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Abstract

Representative samples are important for multivariate calibration. The highly efficient selection of representative samples to be labelled can save money and time. Existing methods, such as Kennard-Stone and net analyte signal selection, are usually based on the distance between candidate samples and labelled calibration sets in feature space. However, these distances are influenced by the feature space, which is spanned by an information vector extracted from labelled samples. To overcome the negative effects of the distance-based selection method, a model performance enhancement-based sample selection method is proposed to select calibration samples efficiently. Based on loss function optimization, the samples that can improve model performance the most, as estimated by bootstrap, are sequentially selected and added to the calibration set. Due to the high representation of each sample, a few samples can build a model that has no significant loss of prediction ability when compared with a model built with the large number set of calibration samples. The performance enhancement-based active learning (PEAL) sample selection method is both effective and efficient.

KEYWORDS

bootstrap modelling, feature space, parsimonious sample selection, performance enhancement, set representation

1 | INTRODUCTION

When applying multivariate calibration of spectroscopic data, it is necessary to select a special subset from a larger calibration set that shall bear the burden of representing the entire relevant data model. In general, such a subset must satisfy two opposing requirements: (1) The subset should be of maximal representativity with respect to the entire set, but (2) the subset should simultaneously be noticeably smaller than the total set so that the cost of building the model is reduced as only the selected samples are submitted to time-consuming chemical analysis.[1] In particular, when new samples are correctly characterized by the applicability domain, the modelling method is found to be much less important; that is, different modelling methods, including complex methods such as deep learning, provide equivalent prediction errors.[2]

The suitable subset requirements result in the contradiction of a large number of labelled samples and the high cost of labelling samples. Unsupervised selection methods (random selection [RS] and Kennard-Stone [KS] selection[3]) are economical but less efficient for analyte-oriented information extraction; as a result, they have low prediction accuracy. Supervised selection (YR[4] and SPXY[5]) requires all samples to be labelled, which is a high monetary cost. Although they

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provide high accuracy, supervised selections are used in a few scenarios. The active learning selection method can learn the information provided by labelled samples sequentially and thus can select representative samples efficiently to reduce label burden significantly.[6]

Based on differences in the data collection procedure, active learning can be divided into pool-based learners and stream-based learners.[7] For stream-based active learning, each unlabelled instance is typically drawn one at a time from the input source, and the learner must decide whether to query or discard it from a stream of input instances. Pool-based active learning queries the most informative instance from a large pool of available unlabelled data. The main difference between stream-based and pool-based active learning is that the former obtains one instance at a time, sequentially, from some streaming data source, and makes each query decision individually. Pool-based active learning, however, evaluates and ranks the entire collection of unlabelled data before selecting the best query.

The most commonly used methods in pool-based selection are committee disagreement (CD) and maximin distance in feature space.[8] In the CD strategy, several parallel calibration models (at least two) are built, the candidate samples are predicted and the samples with the largest variance of different models are submitted to labelling and added to the calibration set. As seen from the procedure, numerous labelled samples are required in the CD method because several models should be built, which significantly reduces the superiority of active learning. The most widely maximin distance methods select samples placed in areas of the feature space not covered by training samples and avoid choosing samples similar to the samples already present in the current training set. The major difference lies in different feature space selections (for all related instance selection methods, not confined to active learning), such as KS (in input space),[3] YR (in output space),[4] SPXY (in integrated input–output space)[5] and NAS (in NAS space).[9] Contrary to pooled data cases, stream-based active learning is suitable for model maintenance and self-adaptation demands to compensate for high system dynamics in online applications.[10] The major task of the stream sample case is representation judgement of a sample determination by latent projection (latent space, which includes T^2 [Hotelling's distance] and Q [standardized residual]).[11] Furthermore, online incremental active learning has progressed to evolving models,[12,13] which are able to expand their knowledge into feature space regions so far unexplored by allowing structural expansions of the models in an incremental manner. Similarly, latent projection space distances are used for sample selection.

Strategies of distance difference in feature space, employed either in a pool-based case or in a stream-based case, are somewhat myopic: They measure only the distance of a single data instance, which is an indirect indicator of performance and opts to be influenced by feature space. As a result, these approaches run the risk of selecting outliers and other poor choices for queries. We can infer that model performance evaluation can give direct instruction on which samples should be selected. As the aim of this paper is to obtain good prediction performance using the least samples, we focus on the pool-based sample selection method.

An adequate number of standards or well-characterized samples are needed in the construction of a robust predictive model, which could represent the global variability that can be expected for analyte concentrations or properties of interest.[14] The range cover requirement guarantees that future predictions involve only the interpolation of the developed calibration model. Forina et al. proposed a sufficient number of samples for partial least square (PLS) modelling of at least 50, preferably 100.[15] Spectra of at least 100 samples must be scanned for a complex solution.[16] We have also studied the sample number required in calibration as general instructions. It is recommended to use the commonly used sample number 100. Gradually adding from 100 to 150 samples, the model accuracy increases almost monotonically. More than 150 samples have little effect on model accuracy.[17] Existing experience shows that the needed samples are usually more than 100, which is somewhat large. The representation and number of calibration sets interact, that is, if the representation of samples is large, then fewer samples are required for modelling.[18] To this end, a more effective selection method that is highly representative is preferred.

In spirit of the above idea, we propose a stepwise sample selection method that can select the most important objects effectively, which results in very few samples being included in the calibration set; however, the stepwise selection method provides no significant loss of prediction ability when compared with a model built with the full set of calibration samples. The presented method has the following strong points: (1) Model performance variation is used as a direct measurement of the representative nature of the sample; as a result, (2) the most representative calibration set can be selected; finally, (3) a small calibration set can be obtained, which is very valuable to reduce the labelling workload.

2 | PERFORMANCE ENHANCEMENT-BASED ACTIVE LEARNING SAMPLE SELECTION METHOD

2.1 | Performance enhanced optimization theory

Let us consider a training set composed initially of n labelled samples $L=\{x_i,y_i\}_{i=1}^{n}$ and an additional learning set composed of m unlabelled samples $U=\{x_u\}_{n+1}^{n+m}$, with $m>>n$. The loss function of the model minimization can be expressed as follows[19]:

$$\widehat{\epsilon}_D=\sum_{i=1}^{n}\ell[f(x_i),y_i]=\frac{1}{2}\sum_{i=1}^{n}(f(x_i)-y_i)^2 \tag{1}$$

where $f(x)$ is the predicted value by the calibration model and y is the real value.

A linear regression model assumes that the regression function $f(x)$ is linear and is parameterized by weight vector θ; then, the model can be formulated as follows:

$$f_\theta(x_i)=\sum_{j=1}^{p}\theta_j x_{ij}=\theta^T x_i \tag{2}$$

where j is the dimension of the data. Substituting Equation 2 into Equation 1 obtains the following:

$$\widehat{\epsilon}_D=\frac{1}{2}\sum_{i=1}^{n}\left(\theta^T x_i-y_i\right)^2 \tag{3}$$

where vector θ represents parameters to be optimized to obtain the minimum $\widehat{\epsilon}_D$.

Assuming that an example x^+ and label y^+ are added to this training set, then the empirical error becomes:

$$\widehat{\epsilon}_D=\sum_{i=1}^{n}\ell[f(x_i),y_i]+\ell_{x^+}[f(x^+),y^+]=\frac{1}{2}\sum_{i=1}^{n}\left(\theta^T x_i-y_i\right)^2+\frac{1}{2}\left(\theta^T x^+-y^+\right)^2 \tag{4}$$

To find a new θ that minimizes the error, θ should be updated according to the negative gradient direction of the loss function:

$$\theta\leftarrow\theta-\frac{\partial\ell_{x^+}(\theta)}{\partial\theta} \tag{5}$$

The respective updating rate $(\partial\ell_{x^+}(\theta)/\partial\theta)$ will be produced for every sample. The sample x^* that makes the largest change in the current model (generating the largest gradient value) should be selected into calibration set:

$$x^*=\operatorname{argmax}\left\|\frac{\partial\ell_{x^+}(\theta)}{\partial\theta}\right\|,x^+\in U \tag{6}$$

For each unlabelled sample x_u, the derivative of the root mean square error $\ell_{x^+}(\theta)$ with respect to θ at x can be expressed by the formula (from the second summand of the right side in Equation 4):

$$\frac{\partial\ell_{x^+}(\theta)}{\partial\theta}=(f(x^+)-y^+)\frac{\partial\left(\theta^T x^+\right)}{\partial\theta}=(f(x^+)-y^+)x^+ \tag{7}$$

In the formula, x^+ is an unlabelled sample in set U. $f(x^+)$ is the predicted value by the built PLS model corresponding to each sample in labelled set L. The exact difficulty lies in y^+ (real value) being unknown.

To estimate the true value of y^+, the bootstrap method[20] is used. Bootstrapping is a nonparametric resampling technique that can be used to estimate the statistical parameters of some populations. Bootstrapping is based on a repetitive estimation of some statistics by changing the population individuals used to estimate that statistic, in this case, expectation. Therefore, the final sample formula in linear regression is as follows:

$$x^*=\operatorname*{argmax}_{x\in U}\frac{1}{K}\sum_{k=1}^{K}\|(f(x)-y_k(x))x\| \tag{8}$$

where K is the number of bootstrap runs and $y_k(x)$ are the prediction values of x for every model bootstrap run generated.

2.2 | Selection procedure

A typical active learning process can be briefly described as follows: (1) Generate a base model from a small initial training set. (2) Select examples with a sampling function from a large set of unlabelled data and label them. (3) Add the newly labelled examples to the training set and update the model. This sampling process is repeated until a certain performance expectation is met or a certain labelled sample number is attained.

To finish the first step, unsupervised selection must be used. The reasonable method is the KS method. The KS algorithm aims to cover the multidimensional space in a uniform manner by maximizing the Euclidean distances between the instrumental response vectors of the selected samples. At each stage, the aim is to select the objects so that they are uniformly spaced over the object space. The first two objects are selected by choosing the two objects that are farthest apart. The third object selected is the one farthest from the first two objects.

The performance enhancement-based active learning (PEAL) sample selection procedure consists of the following steps. The generic flow chart of the PEAL approach is shown in Figure 1.

1. A small batch of datasets $D = \{x_i, y_i\}_{i=1}^{n}$ in the data pool is selected using the KS method (details will be given in Section 2.3) and labelled, that is, response measured.
2. Train the full PLS model using the labelled dataset. We use a full description because all labelled samples are included in the model.

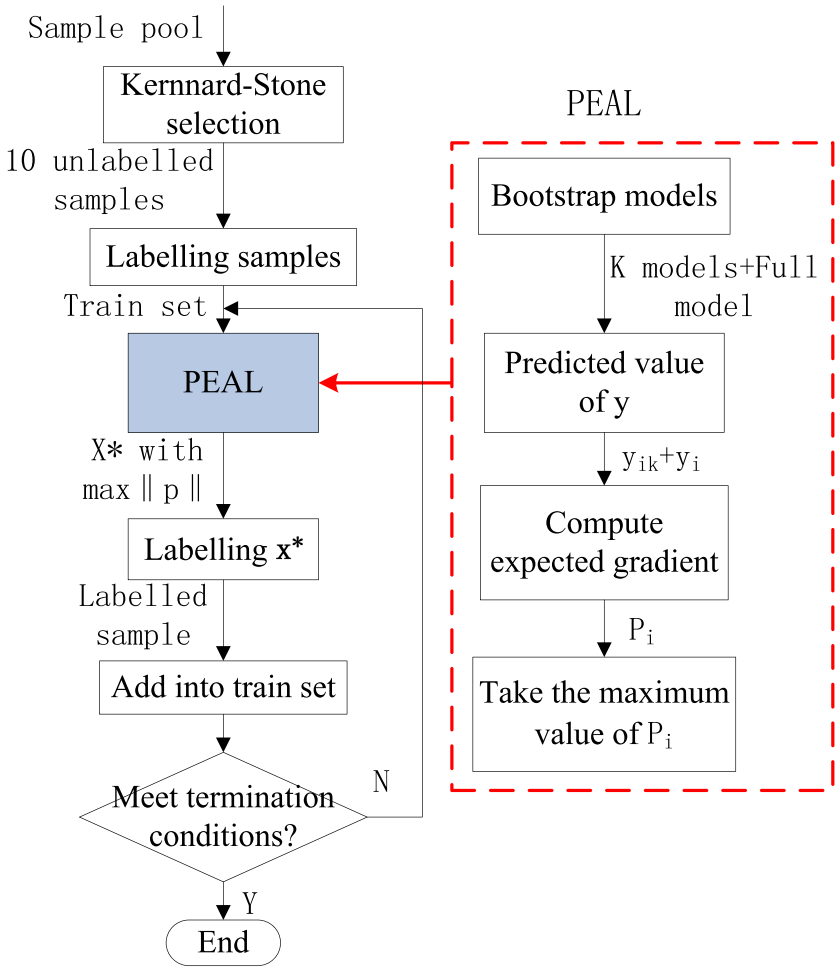


FIGURE 1 Procedures of performance enhancement-based active learning (PEAL) sample selection

3. Construct K (4, as we use, the better, but more computation time is needed) bootstrap calibration set $B(K) = \{f_1, f_2, ..., f_K\}$ by randomly selecting samples of some per cent (60% as we use) in the labelled set. For every bootstrap set, the PLS model is built.
4. The full model and bootstrap models are used to predict every unlabelled sample X_i in the pool, which produces predicted values y_i and y_{ik} ($k = 1, 2 \ldots K$, $i = n+1, n+2, \ldots n+m$).
5. For every sample, compute the expected performance gradient:

$$P_i = \left(y_i - \frac{1}{K}\sum_{k=1}^{K} y_{ik}\right)x_i \tag{9}$$

6. Compute the norm of P for every sample and select the sample $x^* = \text{argmax}\|P\|$.
7. Label x^* and add the sample to the calibration set while deleting the sample from the unlabelled pool.
8. Repeat steps (2) to (7) until the performance enhancement reaches a threshold.

2.3 | KS selection algorithm

KS has been widely used in quantitative spectroscopy and has shown good performance in terms of calibration sampling. KS is a deterministic sequential approach that attempts to select samples uniformly distributed in the predictor space. The KS procedure to select a training or calibration subset of n samples $\left(X_{\text{tr}} = \{x_{\text{tr}j}\}_{j=1}^{n}\right)$ from a given set of N samples $\left(\mathbf{X} = \{x_i\}_{i=1}^{n}\right.$, note that $n < N$) consists of the following:

1. Find in $\mathbf{X}$ the sample $x_{\text{tr}1}$ that is closest to the mean (μ), allocate it in X_{tr} and remove it from $\mathbf{X}$.
2. Find in $\mathbf{X}$ the sample $x_{\text{tr}2}$ that is the most dissimilar to $x_{\text{tr}1}$, allocate $x_{\text{tr}2}$ in X_{tr} and remove it from $\mathbf{X}$.
3. Find in $\mathbf{X}$ the sample $x_{\text{tr}3}$ that is the most dissimilar to those already allocated in X_{tr}. Allocate $x_{\text{tr}3}$ in X_{tr}, and then remove it from $\mathbf{X}$. Note that the dissimilarity between x_{tr} and each x_i is given by the minimum distance of any sample allocated in X_{tr} to each x_i.
4. Repeat step (3) $n - 4$ times to select the remaining samples $(x_{\text{tr}4}, \ldots x_{\text{tr}n})$.

For distance computations in the KS algorithm, the Euclidean distance is commonly used.

3 | EXPERIMENTS

3.1 | Sample collection and analyte measurement

One set of corn gluten meal had 878 samples, and another set of soy sauce had 786 samples collected from a factory and analysed. Two hundred samples were randomly selected from the dataset as the calibration set, whereas 50 samples were selected as the test set. For corn gluten meal, moisture and protein are modelled. For soy sauce, total acid (TA) and ammonia acid nitrogen (AN) are analytes. A reference analysis of TA and AN was performed by titration using NaOH and methanol standard solutions, respectively. The moisture content was measured indirectly through dry mass determination at 105°C, according to GB/T 12087-2008. The protein content was measured by the Kjeldahl method described in GB 5009.5-2010 using an NYK6160 analyser (Shanghai Yihon Analysis Instruments Company, Shanghai, China). The statistical parameters of the calibration analyte are given in Table 1.

TABLE 1 Statistical parameters of dataset

Analyte	Min	Max	Mean	Std
Moisture	5.1	10.9	7.8523	1.0135
Protein	44.4	62.8	57.7707	3.4595
AN	0.3	0.88	0.5920	0.0742
TA	0.46	1.52	0.9392	0.1545

Abbreviations: AN, ammonia acid nitrogen; TA, total acid.

3.2 | Spectral measurements and preprocessing

Spectral data for the soy sauce samples were collected by transflectance in the near infrared region of 950–1650 nm in 2-nm increments using a diode-array analyser (DA 7200, Perten Instruments, Sweden). Spectral data for corn gluten meal were measured by reflectance in 5-nm increments using the same type of spectrometer and wavelength region. Duplicates of each sample were scanned twice. The average spectrum of each sample was used in the subsequent data analysis.

Typical spectra of soy sauce and corn gluten meal are shown in Figure 2A,B, respectively. Figure 2B shows distinct baseline shifts in the raw spectra of the corn gluten meal set. This phenomenon is attributed to scattering due to different powder particle sizes. The baseline shifts were eliminated by a standard normal variate (SNV) preprocessing method. Mean centring was performed for the calibration spectra, and the mean vector was subtracted in the test set. PLS is used in the modelling set-up.

4 | RESULTS AND DISCUSSION

4.1 | Set representative test

Whenever some samples are extracted from a larger population of samples, the representativity of the extracted set towards the original population should be achieved. Two statistical tests are used to compare two datasets and estimate their representativity. The first statistical test is the comparison of the variance–covariance matrices of the two datasets: Their equality implies that both datasets have the same direction in space and that the spread of the data points around the mean is similar. Then, the Mahalanobis distance between the centroids of the two sets is calculated to determine whether the centroids have the same position. The computation details are provided by Jouan-Rimbaud et al.[21] These tests, when applied together, can be used as a diagnostic for the determination of representativity. To avoid the collinearity effect of wavelength, principal components (PCs) are used as variable dimension p in the similarity test. The results of representativity tests of the selected calibration set and test set to the whole set are given in Table 2. Parameter C is a measure of direction similarity, and D^2 is the distance between two centroids. Both significance test

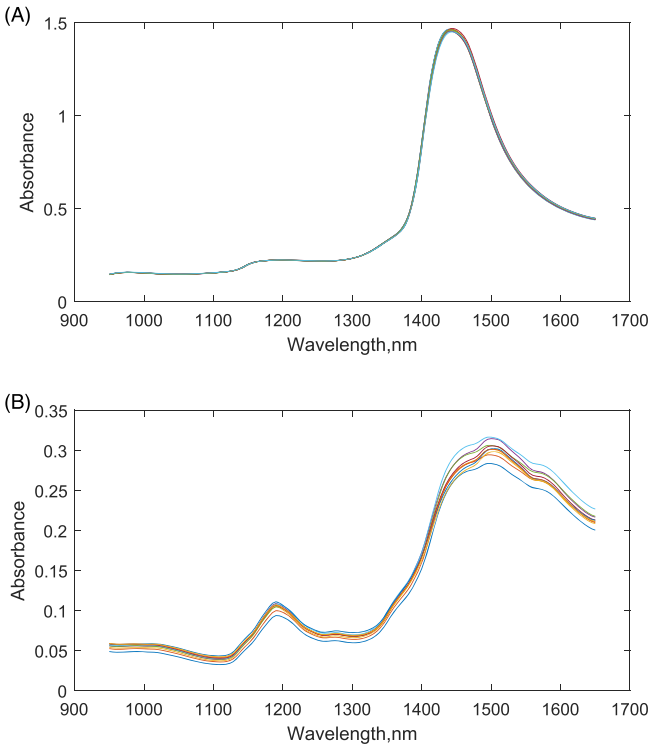


FIGURE 2 Typical spectra of (A) soy sauce and (B) corn gluten meal

TABLE 2 Representative test of the full set to the PEAL set and the full set to the test set

	C	C_{crit}	D^2	D^2_{crit}
Full PEAL	3.3631	25.0000	0.0033	2.21
Full test	13.3266	25.0000	0.0529	2.21

Abbreviation: PEAL, performance enhancement-based active learning.

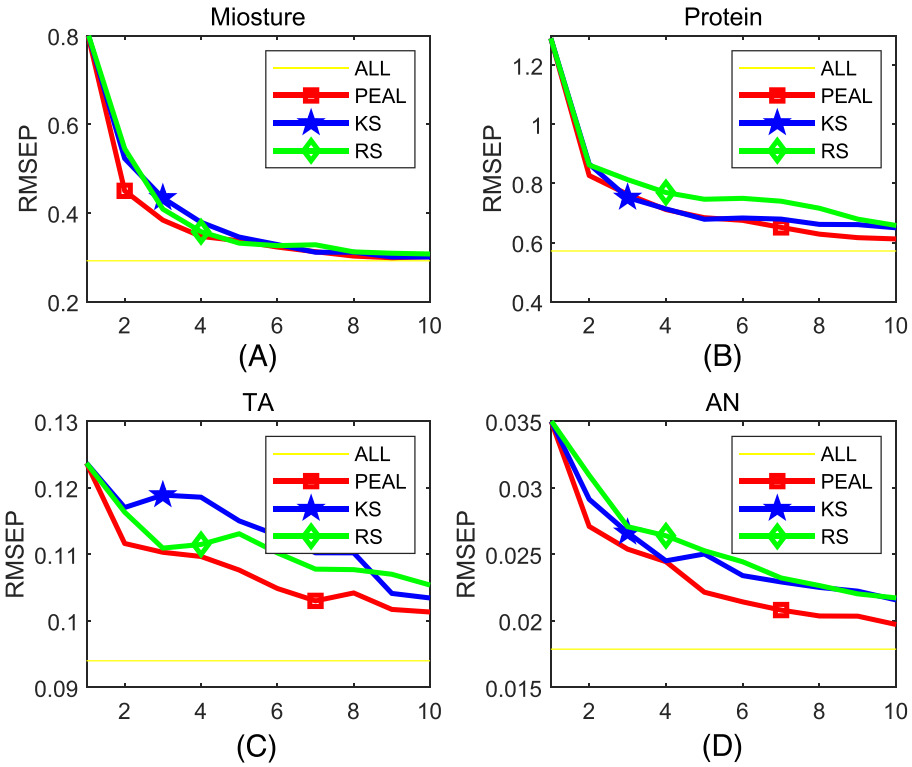


FIGURE 3 Error trend with samples added stepwise using different selection methods for (A) moisture, (B) protein, (C) AN and (D) TA. AN, ammonia acid nitrogen; KS, Kennard-Stone; PEAL, performance enhancement-based active learning; RS, random selection; TA, total acid

parameters are smaller than the threshold value (the degree of freedom is greater than 20, so the threshold of infinite degrees of freedom is uniformly taken, i.e., C_{crit} is 25.000 and D^2_{crit} is 2.21), which means that both the calibration and test sets are representative of the whole dataset.

4.2 | Experimental results

Ten initial samples are selected using the KS method, and then one sample is added every step. RS and KS selection were also performed for comparison. The model performance with calibration number for different selection methods is plotted in Figure 3. PEAL obviously performs better than RS and KS for all analytes. Then, details can be observed for different contents.

The moisture reaches stable accuracy for approximately 20 samples (10 actively selected samples, adding 10 bases), and the error decreases slowly. Furthermore, all selection methods provide good performance, which is due to the predominant absorbance of water, but the prediction error of PEAL decreases the fastest among all selection methods, which means that PEAL selects the most informative samples. Overall, the moisture content can be easily measured by near infrared, and a few samples are required to establish a robust calibration model. The error curve of protein decreased less abruptly than the error curve of moisture. The decrease becomes flat when the iteration reaches 20 samples and consistently decreases slowly. The advantages of PEAL are conspicuous for TA and AN. The profile of

TA was almost lower than the profile of RS and KS in the whole selection procedure. As shown in Figure 3C,D, PEAL not only has a better decline rate than KS and RS but also has a significantly lower root mean square error of prediction (RMSEP).

The RMSEPs of the models, both built using 50 samples selected by different methods and all samples (200), are listed in Table 3, showing that the model built with all samples has the lowest RMSEP, which is expected because the sample size is much larger than the chosen sample size. Table 3 shows that the PEAL method is superior to RS and KS in most cases. Only the RMSEP of the moisture component RS was slightly lower than the RMSEP of PEAL, but the difference was only 0.0029, which can be omitted.

4.3 | Sample scatter comparison between PEAL and KS selection

Sample scatter of PEAL and KS are given in space combined by the first principal component of spectra (PC1) and concentration of analyte (y), protein as we used. Full samples are represented by circles; 10 samples selected in the first step are represented by diamonds; the PEAL-selected samples are marked by triangles; and KS samples are marked by pentagrams, as shown in Figure 4. The selection strategy of PEAL is illustrated through a visual scatter diagram. The PEAL-selected samples are basically evenly distributed both in the X and Y, whereas the KS-selected samples are confined mainly to the uniform distribution of X. Because only the distance of sample space X is considered, the KS method cannot obtain samples with higher Y information. Therefore, the validity of sample selection by the PEAL method is well proven.

4.4 | Generalization ability test by several selection runs

To determine whether the high selection efficiency can be generalized for different sample sets, 10 selection procedures are performed by randomly selecting the full calibration set (200) and test set (50) with every run from the entire

TABLE 3 RMSEP of the model using 50 samples selected by different methods and all 200 samples

	RS	KS	PEAL	All
Moisture	**0.2949**	0.3160	0.2978	0.2922
Protein	0.5243	0.5858	**0.5206**	0.4928
TA	0.0232	0.0270	**0.0228**	0.0219
AN	0.1192	0.1142	**0.1086**	0.1108

Note: The bold values are minimum values of different sample subsets.

Abbreviations: AN, ammonia acid nitrogen; KS, Kennard-Stone; PEAL, performance enhancement-based active learning; RS, random selection; TA, total acid.

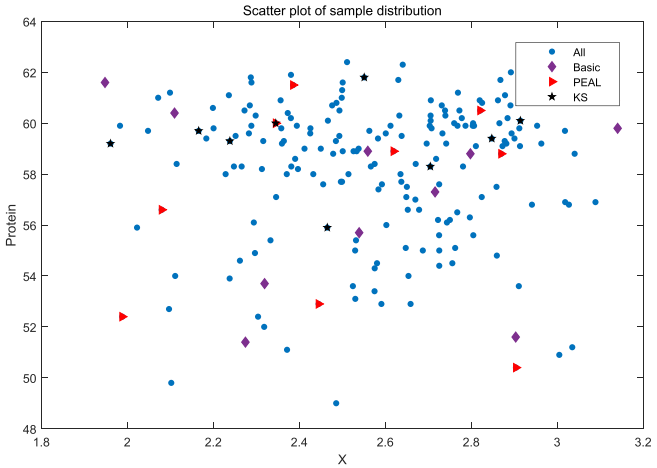


FIGURE 4 Scatter plot of sample distribution by performance enhancement-based active learning (PEAL) and Kennard-Stone (KS)

TABLE 4 Relative error of RS, KS, PEAL set and full set for different analytes

	RS	KS	PEAL
Moisture	0.0432	0.0486	**0.0290**
Protein	0.1524	0.1416	**0.0856**
TA	0.2069	0.4930	**0.1018**
AN	0.1150	0.1566	**0.0640**

Note: The bold values are minimum values of different sample subsets.
Abbreviations: AN, ammonia acid nitrogen; KS, Kennard-Stone; PEAL, performance enhancement-based active learning; RS, random selection; TA, total acid.

sample set (879 corn gluten meal and 786 soy sauce samples) separately. Models built by different calibration sets are used to predict the test set, and the corresponding RMSEPs are obtained. The relative error $\left(\delta = \left[\sum_{i=1}^{10}(e_{si} - e_{ai})/e_{ai}\right]/10\right)$ (e is RMSEP for different calibration sets, subscript s means selected, a means all) for different analytes is listed in Table 4. The largest error value comes from TA, which is 10%, and the other contents are only several per cent of the full set. We can conclude that the performance difference of PEAL selected 50 and full 200 samples is rather small, which means that with the tolerance of a small performance decrease, a large part of label burden can be saved, with higher performance than KS and RS selection of 50 samples in most cases.

5 | CONCLUSIONS

An effective and efficient sample selection method, PEAL, is proposed in this paper. The method evaluates model performance for each sample addition, which provides a more direct indicator for each sample and can improve the model more efficiently. Compared with the commonly used selection method, PEAL can select the most representative sample and achieve stable performance very rapidly, thus reducing the cost requirement of sample labelling. Furthermore, as the model is built by a small number of samples, there will be fewer interference factors included. As a result, the model is more robust, which is very important for long running of the model.

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DATA AVAILABILITY STATEMENT
The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available because of privacy or ethical restrictions.

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